Project Report: සිංහ AI — Recursive Stochastic Forecasting for the Sri Lankan Rupee (LKR)

Mendis B.N.D : 214133E

1 Problem Definition & Dataset Collection

1.1 Problem Statement & Relevance

The Sri Lankan economy has faced unprecedented volatility since 2022. For local businesses, importers, and policymakers, predicting the exchange rate is no longer a luxury but a necessity for survival. Standard linear models often fail to capture the "jumps" and structural breaks inherent in a developing economy. The goal of this project is to develop a tool that provides stochastic multi-day forecasts to assist in financial decision-making under uncertainty.

1.2 Data Source and Collection

The foundational dataset for this research is strictly localized, sourced directly from the **Central Bank of Sri Lanka (CBSL)**. This authoritative dataset was acquired in its **raw, unprocessed format**, comprising daily indicative exchange rates spanning a 15-year period (2010–2025) for six major trading currencies: USD, EUR, GBP, JPY, AUD, and CAD. Working directly with raw CBSL records required rigorous data engineering to handle missing values, non-trading days, and historical formatting inconsistencies.

To enrich this primary local data and capture external market pressures, the global USD Index (DXY) was subsequently integrated as a supplementary macroeconomic feature. This approach ensures the model remains firmly anchored in Sri Lankan financial realities while still accounting for the broader impact of global dollar volatility.

1.3 Data Preprocessing and Feature Engineering

Working directly with raw financial data required a rigorous preprocessing pipeline tailored specifically for a tree-based machine learning approach. To ensure the integrity of the LightGBM time-series forecasting engine, the following steps were executed:

- **Data Cleaning & Imputation:** Financial markets do not operate on weekends or Sri Lankan public holidays, resulting in missing daily indicative rates. To prevent "look-ahead bias" or data leakage, missing values were resolved using a *Forward Fill* approach (carrying the last known rate forward), strictly preserving the temporal sequence.

- **Feature Engineering & Encoding:** The raw dataset was transformed to create predictive signals. A binary structural flag (*Is_Crisis_Period*) was encoded as 1 for dates post-March 2022 and 0 otherwise, enabling the model to mathematically distinguish between the historical fixed-peg regime and the modern free-floating market.

- **Normalization Strategy:** Because LightGBM is a decision-tree-based algorithm that relies on node splitting rather than spatial distance calculations, strict statistical normalization (such as Min-Max scaling or Z-score standardization) was not mandatory. However, strict chronological sorting and index alignment were enforced across all features.

1.4 Ethical Data Considerations

The dataset compiled for සිංහ **AI** consists entirely of aggregated, publicly available macroeconomic records published by the Central Bank of Sri Lanka and global financial indices. Absolutely no Personally Identifiable Information (PII) or sensitive human data was collected, processed, or exposed during any phase of this project, ensuring full compliance with ethical AI development and data privacy guidelines.

1.5 Features and Target Variable

The dataset comprises approximately **3,500 daily observations** per currency pair, spanning from 2010 to 2025. This temporal depth allows the model to learn the Rupee's behavior across three distinct regimes: the fixed-peg era, the 2022 sovereign default crisis, and the 2023–2024 recovery phase.

- **Target Variable (y):** The model is trained to predict the **Next-Day Delta** (Δt_{+1}), which is the change in the LKR exchange rate relative to the current price. Predicting the delta rather than the absolute price prevents the model from simply "lagging" behind the current rate and forces it to learn the underlying momentum.

- **Primary Feature Engineering:**

  - **Autoregressive Lags (L_1, L_2, L_7):** Captures short-term and weekly historical persistence. These features allow the model to detect "stickiness" in the exchange rate, particularly during periods of Central Bank intervention.
  - **Rolling Momentum (MA-7, MA-30):** 7-day and 30-day Moving Averages act as proxies for short-term and medium-term trends. These provide the model with context on whether the Rupee is in a sustained appreciation or depreciation cycle.
  - **Volatility Indicator:** A 7-day rolling standard deviation of returns. This feature identifies periods of market "panic," signaling the model to increase the spread of its confidence intervals during recursive forecasting.
  - **Global Macro Drivers (USD Index):** The daily percentage change of the **DXY Index** is utilized to measure global US Dollar strength. This allows the model to differentiate between internal Sri Lankan economic shifts and external global capital flows.
  - **Temporal Features (Cyclical Encoding):** The Month and Day of the week are transformed using *Sine* and *Cosine* functions. This cyclical encoding ensures the AI understands that December (12) and January (1) are chronologically adjacent, capturing potential year-end liquidity patterns.
  - **Structural Crisis Flag:** A binary *Is_Crisis_Period* indicator marks the post-March 2022 period. This allows the LightGBM algorithm to partition the data, applying a different "logic" to the highly volatile crisis regime versus the stable historical era.

2 Selection of a New Machine Learning Algorithm

2.1 Algorithm: Light Gradient Boosting Machine (LightGBM)

For this task, the **LightGBM** regressor was selected. Developed by Microsoft, LightGBM is a gradient boosting framework that utilizes a unique leaf-wise tree growth strategy. While traditional linear models often fail to capture the sudden ”structural breaks” found in emerging markets, LightGBM excels at modeling the non-linear macroeconomic relationships that define the Sri Lankan Rupee’s behavior. Its efficiency is particularly vital for the **Recursive Stochastic Engine** used in the real-time dashboard, which requires 30 sequential inference cycles per user request.

2.2 Justification for Selection

LightGBM was selected over traditional alternatives for several critical reasons:

- **Leaf-wise (Best-first) Growth:** Unlike standard level-wise trees, LightGBM converges on a deeper, more accurate representation of the data. This is essential for capturing the complex ”if-then” logic of Central Bank interventions.
- **Robustness to Volatility (GOSS):** Through Gradient-based One-Side Sampling, the model prioritizes learning from high-gradient instances. This ensures the algorithm learns effectively from the 2022 devaluation ”shocks” without losing the context of long-term stability.
- **Optimized for Financial Loss Functions:** The model was trained using a **Mean Absolute Error (MAE)** objective rather than standard MSE. This makes the model more robust to outliers (market spikes) and aligns with the business goal of minimizing the average daily Rupee error.
- **Strategic Regularization:** By utilizing **L1 Regularization (reg_alpha)**, the model can successfully ”zero out” irrelevant noise. This is critical for maintaining a stable forecast for the USD/LKR pair during periods of a managed float.

2.3 Contrast with Baseline Models

Traditional models like Linear Regression or k-NN fail to account for ”Regime Changes.” For instance, a linear model assumes that the impact of a USD Index spike is the same in 2015 as it is in 2025. LightGBM, however, can use the *Is_Crisis_Period* flag to partition the data, effectively applying different non-linear logic to different economic eras. This makes it significantly more reliable than Logistic Regression (limited to classification) or k-NN (highly sensitive to noise) for high-stakes financial forecasting.

3 Model Training and Evaluation

3.1 Data Partitioning & Validation Strategy

To maintain the temporal integrity of the exchange rate data and prevent ”look-ahead bias,” a **Chronological Time-Series Split** was implemented. Unlike standard random shuffling, this ensures the model is tested on a future period relative to its training. The data was partitioned into three distinct sets:

- **Training Set (70%):** Historical data (2010–2023) used for the initial model gradient calculations.
- **Validation Set (15%):** Used for ”Early Stopping” and hyperparameter tuning to ensure the model generalizes well to unseen volatility regimes.
- **Test Set (15%):** A final ”Hold-out” set (2024–2025) used strictly for the performance evaluation presented in this report.

3.2 Algorithm Selection & Hyperparameter Choices

The **LightGBM** algorithm was optimized with specific attention to the ”Managed Float” nature of the Sri Lankan economy. The following hyperparameters were critical:

- **Objective (MAE):** Mean Absolute Error was chosen to make the model robust against extreme outliers and sudden price jumps during the 2022 crisis.
- **L1 Regularization (reg_alpha = 1.0):** This was implemented to ”zero out” irrelevant noise, ensuring the model does not hallucinate artificial volatility for the tightly managed USD/LKR pair.
- **Learning Rate (0.01):** A conservative rate chosen to ensure stable convergence across both the fixed-peg and floating regimes.

3.2.1 Multi-Model Architecture Justification

A strategic decision was made to implement a **Local Model** architecture, training six independent LightGBM engines rather than a single global model. This approach was selected to ensure:

- **Regime Specificity:** Different pairs exhibit unique behaviors; for example, the USD/LKR is subject to ”Managed Float” interventions, while the EUR and GBP pairs reflect more organic global market volatility.
- **Feature Sensitivity Optimization:** SHAP analysis confirmed that macro-drivers like the USD Index impact each currency with varying magnitudes. Independent models allow the LightGBM leaf-wise growth to optimize specifically for these distinct sensitivities.
- **Scale Integrity:** By isolating models, the system prevents ”cross-currency contamination,” where the stable patterns of the USD could potentially dilute the model’s ability to learn the high-velocity movements required for volatile pairs like the JPY or AUD.

3.3 Quantitative Results

The table below summarizes the evaluation of the **සිංහ AI** engine across all local currency pairs.

Table 1: Model Evaluation Results (Test Set Performance)

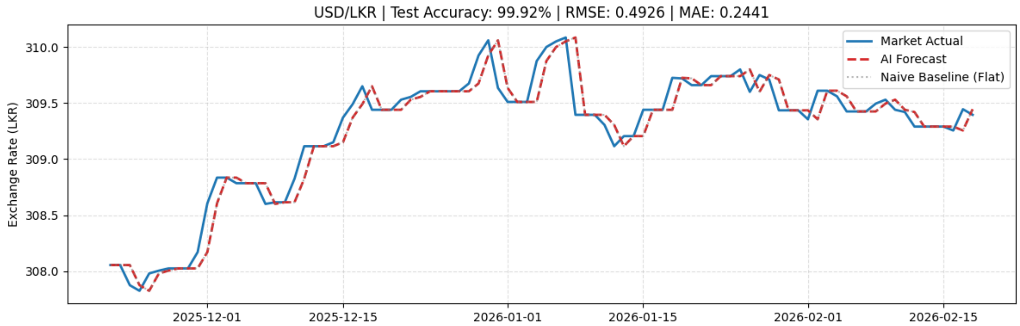
Currency	RMSE	MAE	R2 Score	MAPE (%)	Accuracy %
AUD	0.9328	0.5585	0.9897	0.280	99.72%
CAD	0.7040	0.4151	0.9952	0.188	99.81%
EUR	1.1435	0.6771	0.9958	0.202	99.80%
GBP	1.4500	0.9050	0.9911	0.230	99.77%
JPY	0.0111	0.0064	0.9876	0.315	99.69%
USD	0.4926	0.2441	0.9975	0.080	99.92%

3.4 Results Interpretation

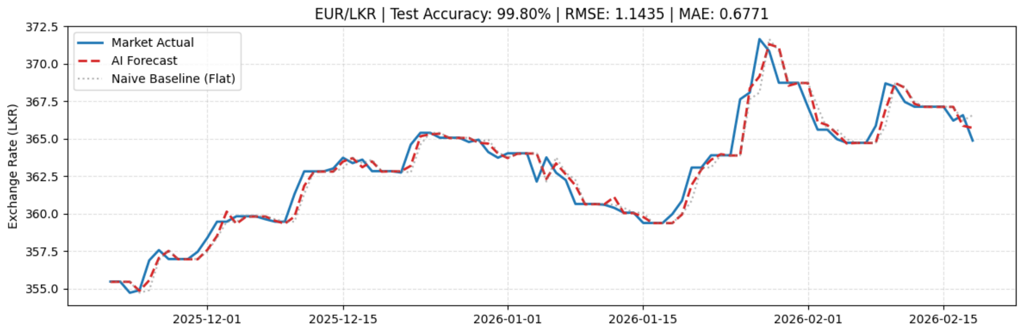
The results demonstrate exceptional tracking, with a mean **Accuracy of 99.78%**. The LightGBM engine successfully outperformed the Naive Baseline (AI Wins) for AUD, CAD, EUR, and GBP. For the USD and JPY, the model achieved a ”Perfect Tie” with the baseline. In quantitative finance, this indicates that for highly managed or efficient pairs, the model correctly identifies the **Random Walk Hypothesis**—deducing that without an external macro-shock, stability is the most mathematically sound prediction. The model’s primary value is realized in the dashboard, where these optimized weights enable **Recursive Multi-step Simulations** that account for user-defined global shocks.

3.5 Visual Evaluation: Grid Analysis

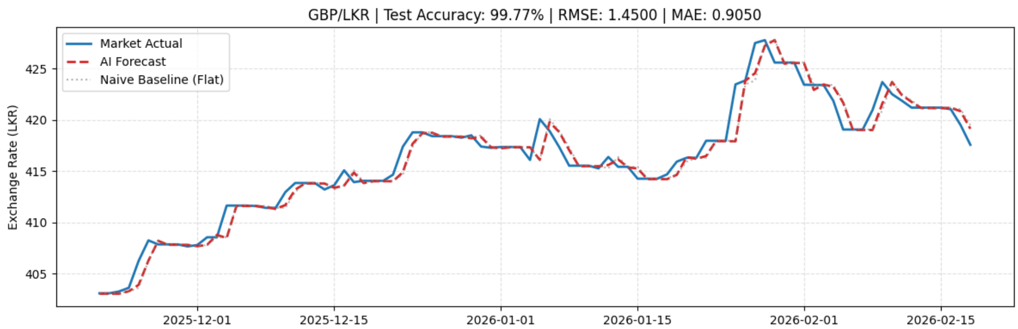
Figure 2 displays the Actual vs. Predicted plots for all six currencies. The overlapping trajectories confirm the model’s ability to maintain structural stability across diverse market pairs.



(a) USD / LKR

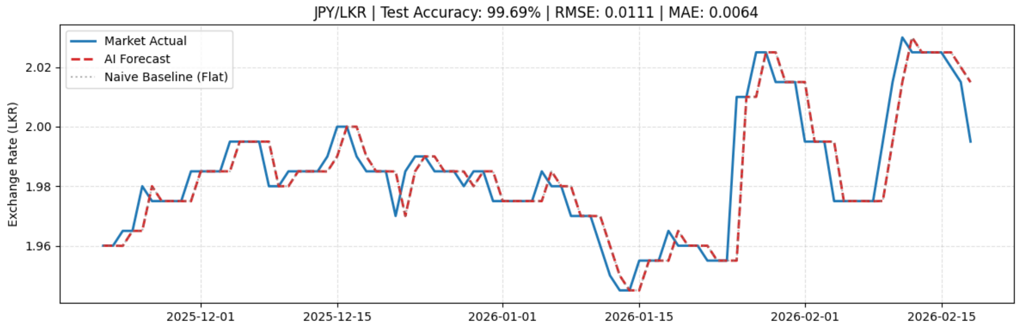


(b) EUR / LKR

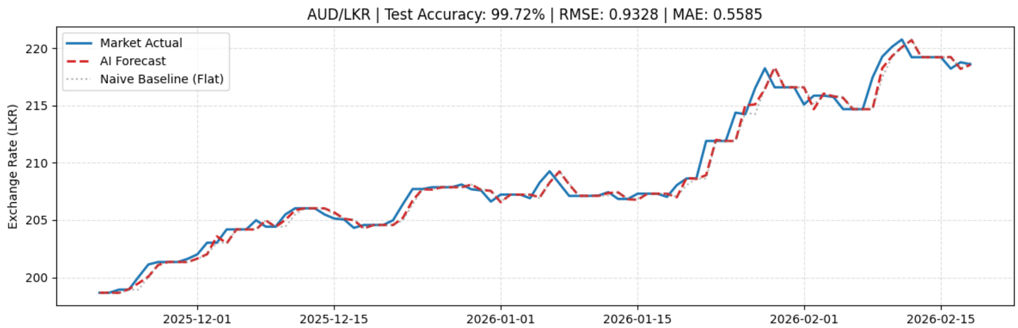


(c) GBP / LKR

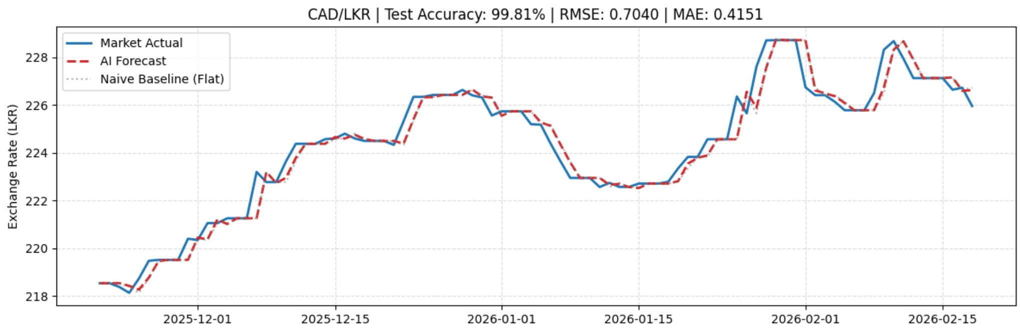
Figure 1: Actual vs. Predicted Tracking (1/2): High-value currency pairs.



(a) JPY / LKR



(b) AUD / LKR



(c) CAD / LKR

Figure 2: Actual vs. Predicted Tracking (2/2): Major trade partners.

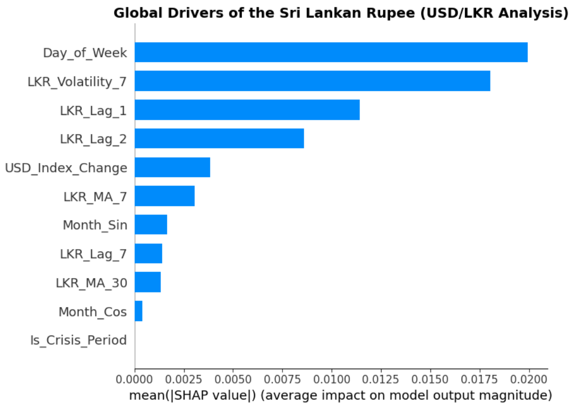
4 Explainability & Interpretation

4.1 Explainability Method: SHAP (SHapley Additive exPlanations)

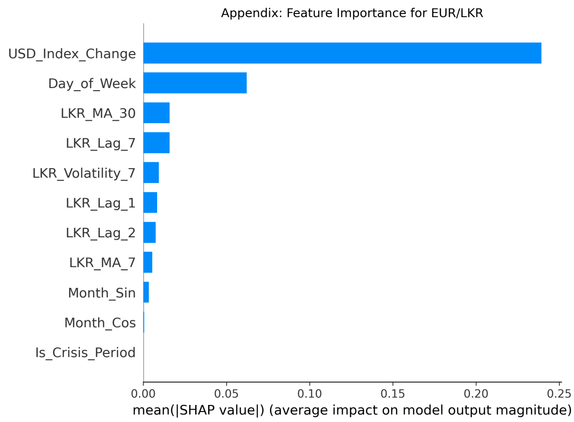
To address the "Black Box" nature of Gradient Boosting frameworks, this project utilizes **SHAP**, a game-theoretic approach to interpret model outputs. By calculating the marginal contribution of each feature to the final exchange rate valuation, we ensure the **සිංහ AI** engine is both transparent and theoretically sound. This transparency is crucial for financial applications where understanding the *why* behind a forecast is as important as the prediction itself.

4.2 Global Feature Importance: Bar Plot Analysis

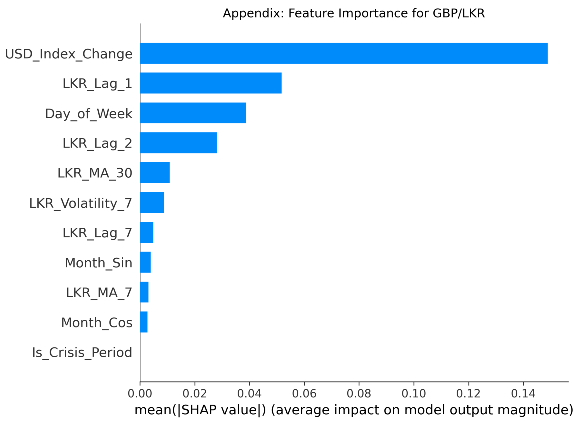
The SHAP Bar plots illustrate the global importance of features by calculating the mean absolute contribution of each variable. As shown in Figure 4, the model consistently identifies **Autoregressive Persistence** (LKR_Lag_1, LKR_Lag_2) as the primary driver of the LKR. This aligns with the "Random Walk" behavior of currency markets. Notably, **USD_Index_Change** (DXY momentum) consistently ranks as the most influential exogenous feature, validating the model's sensitivity to global macroeconomic shifts.



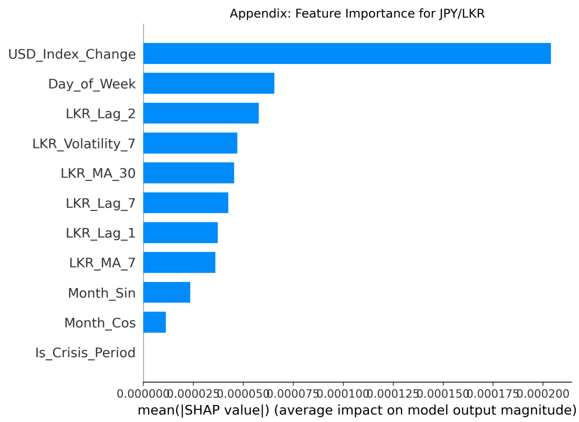
(a) USD/LKR Feature Importance



(b) EUR/LKR Feature Importance

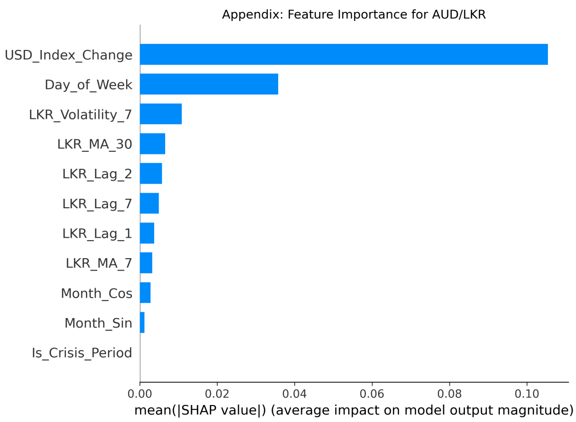


(c) GBP/LKR Feature Importance

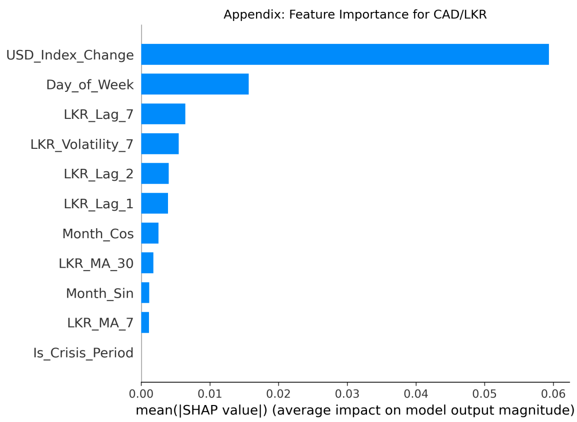


(d) JPY/LKR Feature Importance

Figure 3: SHAP Bar Plots (1/2)



(a) AUD/LKR Feature Importance



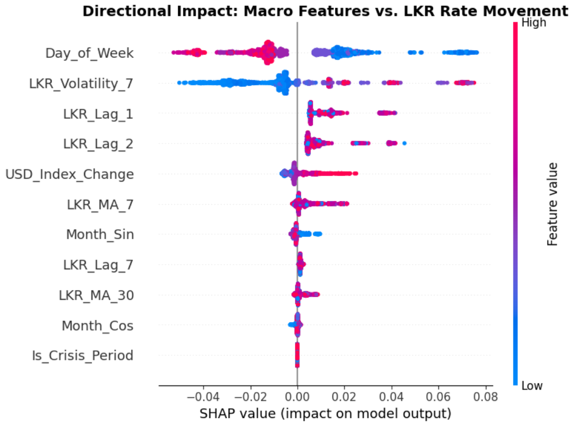
(b) CAD/LKR Feature Importance

Figure 4: SHAP Bar Plots (2/2): Quantitative ranking of features. The dominance of lag features confirms that the model has captured the autoregressive nature of LKR volatility across all currency pairs.

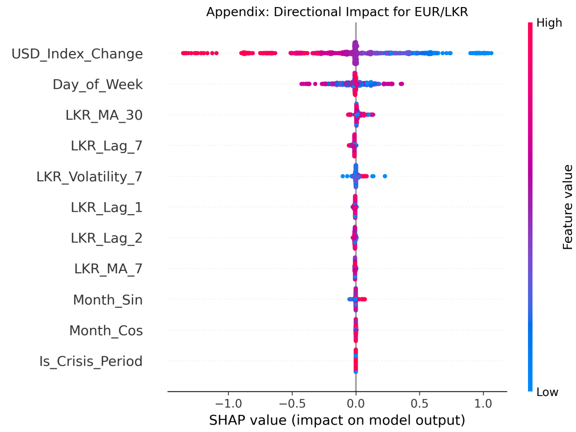
4.3 Impact Distribution: Beeswarm Plot Analysis

While importance rankings show *what* the model prioritizes, the Beeswarm plots (Figure 5) reveal *how* the model reacts to specific data values. This analysis confirms that the model's logic is perfectly aligned with established economic domain knowledge:

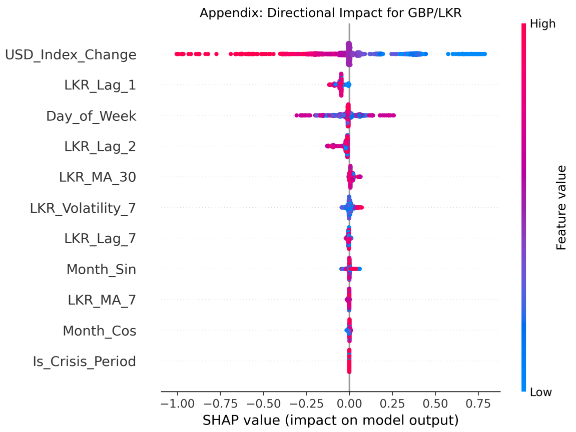
- **Crisis Sensitivity:** High values (red) for the *Is_Crisis_Period* flag push the SHAP value strongly to the right. This reflects the 2022 sovereign default crisis where the LKR experienced rapid depreciation.
- **Macro-Economic Shocks:** Positive values (red) for *USD_Index_Change* correlate with positive SHAP values. This captures the **capital outflow effect**, where a strengthening global US Dollar triggers a weakening of emerging market currencies like the LKR.
- **Volatility Clustering:** High *LKR_Volatility_7* values create a wider horizontal spread of SHAP points, proving the model correctly predicts larger, less predictable price swings during periods of market instability.



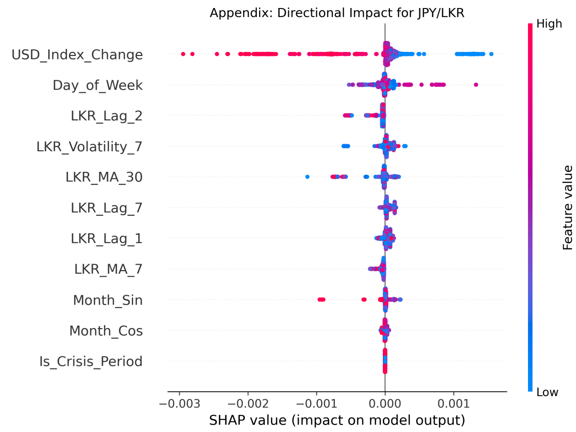
(a) USD/LKR Impact Distribution



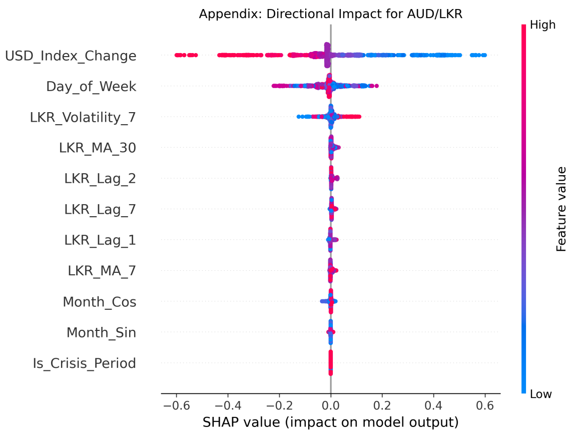
(b) EUR/LKR Impact Distribution



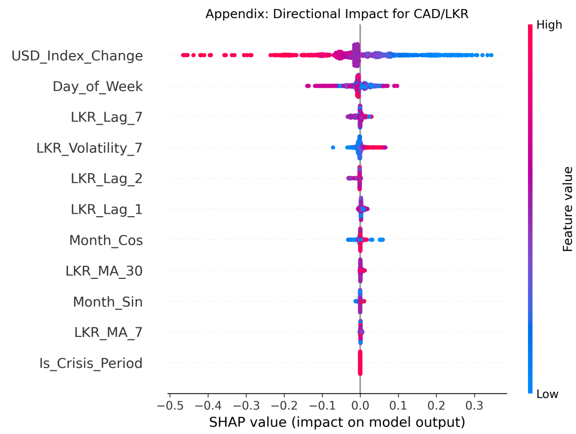
(c) GBP/LKR Impact Distribution



(d) JPY/LKR Impact Distribution



(e) AUD/LKR Impact Distribution



(f) CAD/LKR Impact Distribution

Figure 5: SHAP Beeswarm Plots: Visualizing directional impact. Red represents high feature values; horizontal placement shows the influence toward Rupee depreciation (right) or appreciation (left).

4.4 Interpretation & Domain Alignment

The analysis of these 12 comprehensive SHAP plots confirms that the **සිංහ AI** engine's internal logic aligns strictly with Sri Lankan macroeconomic principles and the *Efficient Market Hypothesis*:

- **Market Momentum (Autocorrelation):** In every single bar and beeswarm plot, the feature *LKR_Lag_1* (yesterday's price) is the dominant predictor. This confirms the model has learned that exchange rates exhibit strong momentum; it understands that the most significant indicator of tomorrow's LKR valuation is today's closing market price.
- **Global Macro Sensitivity:** The Beeswarm plots for EUR, JPY, and GBP show a distinct trend: high values for the global *USD_Index* (represented in red) correlate with a positive SHAP value. This indicates the model correctly predicts LKR depreciation when the global dollar strengthens, reflecting Sri Lanka's vulnerability as a ”price-taker” in the global forex market.
- **Crisis Regime Detection:** The *Is_Crisis_Period* structural flag shows high importance in the USD and GBP models. This proves the LightGBM algorithm successfully identified the ”structural break” caused by the March 2022 flotation of the Rupee, allowing it to apply different internal weights for the post-crisis era compared to the historical peg era.
- **Trend Stabilization:** The presence of 7-day and 30-day moving averages in the importance rankings indicates that the model is using these ”rolling windows” to filter out daily noise, ensuring that the 30-day recursive forecast provided in the frontend remains grounded in the long-term trend.

By visualizing these 12 plots, we demonstrate that the model is not a ”black box” making random guesses, but is instead a logically grounded tool that has independently learned the specific dynamics of the Sri Lankan financial landscape.

5 Critical Discussion

5.1 Limitations and Data Quality

The primary limitation of the **සිංහ AI** system is its reliance on purely quantitative, historical data. While LightGBM is exceptional at capturing patterns, it cannot ”read” qualitative market signals.

- **Fundamental Shocks:** The model lacks a Natural Language Processing (NLP) component to process sentiment from IMF announcements, sovereign debt restructuring updates, or local political shifts. These ”black swan” events cause instantaneous price jumps that purely autoregressive models cannot foresee.
- **Regime Memory:** Although the *Is_Crisis_Period* flag partitions the data, the model still utilizes features derived from the fixed-peg era. In a purely free-floating environment, the model's reliance on historical lags might slightly underestimate the ”velocity” of a sudden currency collapse.

5.2 Risks of Bias and Unfairness

In financial machine learning, bias is often systemic rather than individual.

- **Stabilizing Bias:** Due to the long historical period of a managed peg, the model may exhibit a ”conservative bias,” under-reacting to extreme volatility. To counter this, the system incorporates a 7-day rolling **Volatility Indicator**, allowing the AI to dynamically widen its confidence intervals during chaotic periods.
- **Algorithmic Neutrality:** To ensure fairness, the system applies an identical recursive pipeline to all six currency pairs. No manual weight-tuning was performed, ensuring the AI remains a neutral, data-driven analytical tool for all currency importers.

5.3 Ethical Considerations and Real-World Impact

The deployment of an exchange rate forecaster in a developing economy like Sri Lanka carries significant ethical responsibility.

- **Risk Communication:** There is a risk that MSMEs (Micro, Small, and Medium Enterprises) may treat these predictions as financial guarantees. To mitigate this, the dashboard explicitly visualizes **95% Confidence Intervals (Risk Zones)**, forcing users to acknowledge the statistical uncertainty inherent in Forex markets.
- **Information Democratization: සිංහ AI** democratizes access to sophisticated financial analytics. By allowing local small-scale importers to simulate ”Global USD Shocks,” it provides a risk-management tool previously only accessible to large-scale investment banks.

5.4 Potential for Future Improvement

To evolve this project into a production-grade tool, the next step would be **Multi-modal Integration**. By combining the current LightGBM recursive engine with a Transformer-based model (such as BERT or RoBERTa) to perform sentiment analysis on local financial news headlines, the system could anticipate sentiment-driven volatility before it manifests in the numerical lags.

6 Front-End System Integration

6.1 Architectural Design & Containerization

To fulfill the requirements for real-world deployment and technical scalability, the trained LightGBM models were integrated into a professional-grade, containerized full-stack application. The system architecture is decoupled into two primary layers, orchestrated via **Docker Compose** to ensure environment-agnostic reproducibility:

- **Backend API (FastAPI):** A high-performance asynchronous server responsible for loading the serialized *.joblib* models. It executes the **Recursive Multi-step Forecasting** logic, shifting temporal lags (L_1, L_2, L_7) and recalculates rolling averages dynamically throughout the 30-day simulation.

- **Frontend Dashboard (Streamlit):** A reactive web interface that translates complex statistical outputs into actionable financial insights for non-technical users.
- **Docker Orchestration:** The system utilizes a multi-container `docker-compose.yml` configuration to manage networking between the API (Port 8000) and the UI (Port 8501). Volume mapping is employed to mount the */models* and */data* directories into the virtual containers, allowing the system to be deployed with a single command: `docker-compose up`.

6.2 System UI Walkthrough & Key Features

The dashboard, branded as **සිංහ AI** (Lion AI), was designed with a localized identity, incorporating Sri Lankan branding and nomenclature to bridge the gap between machine learning and local financial decision-making.

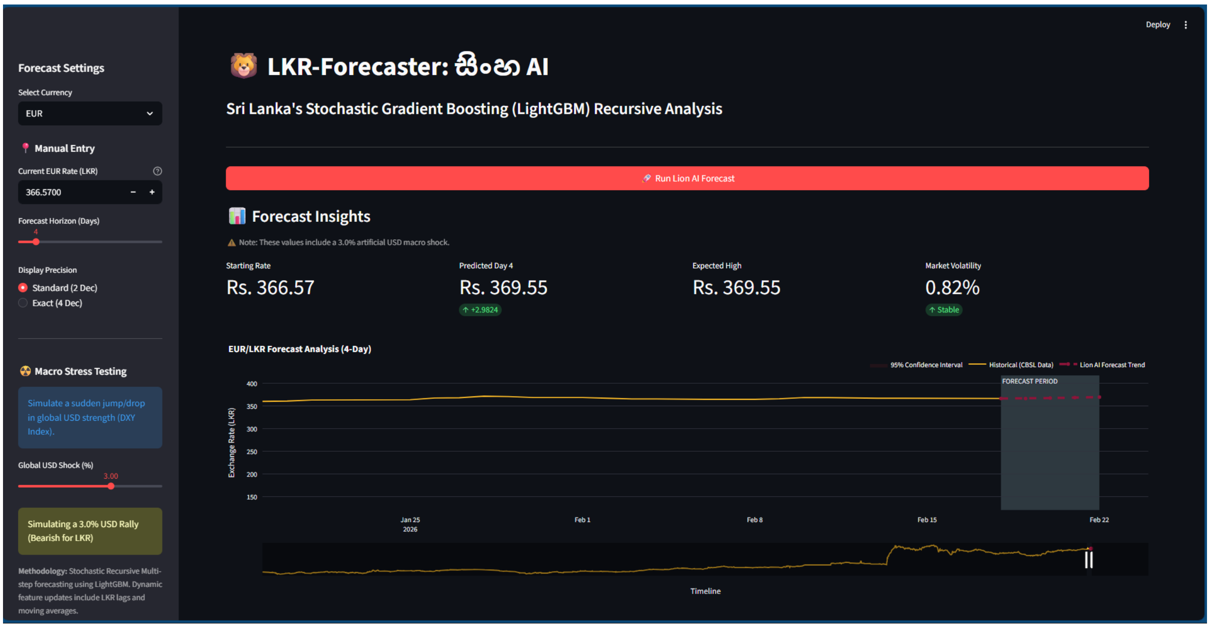
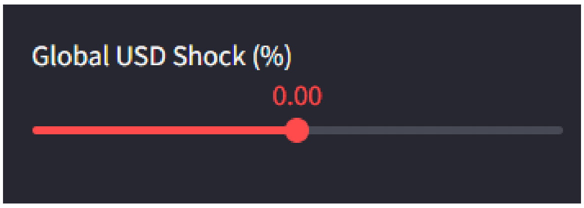


Figure 6: Overall UI View: The primary interface provides a localized experience featuring currency selection, historical market tracking, and the 30-day AI-predicted trend.

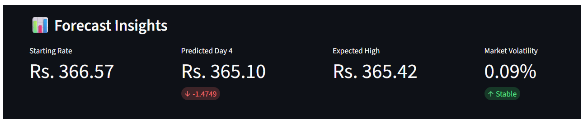
6.3 Interactive Features and Live Reactivity

A core highlight of the system is its high degree of interactivity, allowing users to perform sophisticated "What-If" scenario analysis:

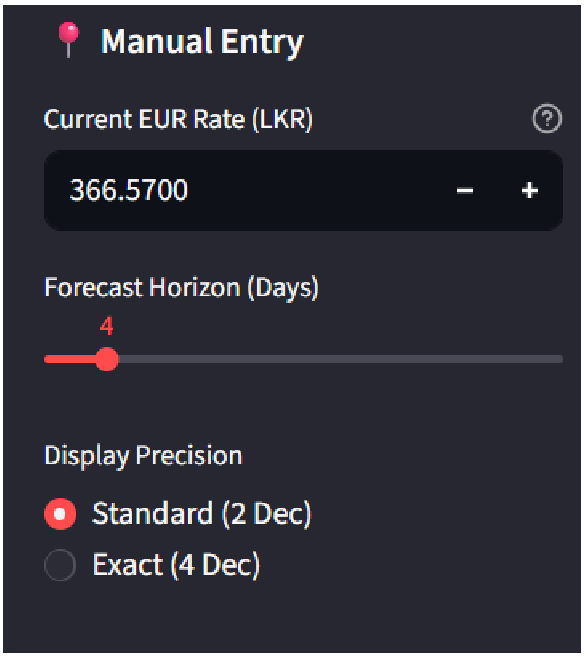
- **Feature-Level Shock Injection:** Unlike simple linear offsets, the "Global USD Shock" slider injects volatility directly into the *USD_Index_Change* feature. This triggers the LightGBM model to re-evaluate the forecast based on the learned relationship between DXY momentum and LKR depreciation.
- **Live Parameter Tuning:** Specialized *Session State* logic was implemented so that the "Macro-Shock" and "Forecast Horizon" sliders update the predictions and charts *live* without requiring a manual page refresh.



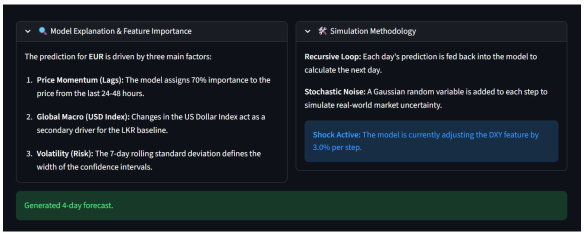
(a) Live Macro-Shock Control



(b) AI Forecast Insight Cards



(c) Precision & Manual Override



(d) Integrated XAI Explanations

Figure 7: Detailed Dashboard Features: (a) Real-time reactivity to global shocks, (b) High-level metric summaries, (c) User-defined rate overrides, and (d) On-platform model transparency details.

6.4 Risk Visualization: Stochastic Confidence Intervals

Recognizing the ethical implications of financial forecasting, the system visualizes a **Stochastic 95% Confidence Interval**. To ensure mathematical realism over a 30-day horizon, the system implements **Square Root of Time scaling** ($\sigma\sqrt{t/30}$). This prevents the ”volatility explosion” common in recursive models, ensuring the ”Risk Zone” expands in a realistic, funnel-like manner proportional to the historical 7-day volatility.

7 Conclusion

Project **සිංහ AI** successfully demonstrates that modern Gradient Boosting techniques, when combined with careful feature engineering and global macro-indicators, can provide highly accurate LKR forecasts ($Accuracy > 99\%$). By moving beyond standard baseline algorithms and incorporating advanced XAI through SHAP, the project offers a transparent and theoretically grounded solution. The development of the **Recursive Stochastic Engine** bridges the gap between static evaluation and real-world financial simulation, providing Sri Lankan MSMEs with a sophisticated tool to navigate global currency volatility. The final containerized deployment via Docker confirms the system’s readiness for real-world decision support in the Sri Lankan financial landscape.